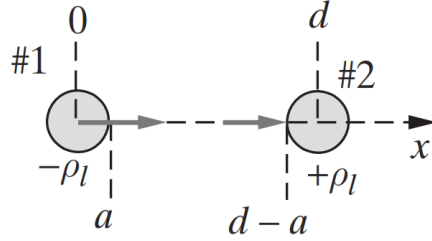


Q1: [40] Consider a **two-wire** transmission line configuration used in telephony, radio engineering, and electric power distribution, consisting of two parallel cylindrical conductors, each of radius a , separated by a distance d . Assume, $d \gg a$ so that the radius of the conductor may be ignored, and the two wires may be considered as the line charge densities. The line charge at $x = 0$ has a density $-\rho_l$, and the line charge at $x = d$ has a density $+\rho_l$.



1. Find the net electric field $\mathbf{E}$ at a distance x between the wires, where $a < x < d - a$.
2. Draw the lines and direction of the electric field in the space between the wires. Also, draw the equipotential lines.
3. Next, integrate the field $\mathbf{E}$ in the range $a < x < d - a$ to find the potential difference $V_{21} = V_2 - V_1$.
4. Using V_{21} , obtain the capacitance between the two wires.

Solution: We know that the electric field at a distance ρ from a line charge of linear charge density ρ_l is given as $E_\rho = \rho_l / (2\pi\epsilon_0\rho)$. Taking the $+\rho_l$ and $-\rho_l$ line charges to be located at $\rho = d$ and $\rho = 0$, respectively, the electric field as a function of distance x along the x -axis (i.e., $y = 0$ and $z = 0$) is given as

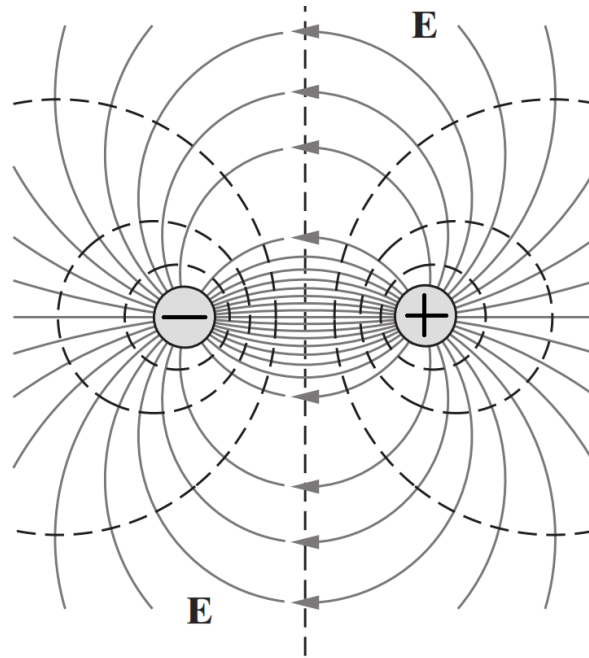
$$E_x(x, 0, 0) = \frac{-\rho_l}{2\pi\epsilon_0 x} + \frac{\rho_l}{2\pi\epsilon_0(x - d)}$$

To find the potential difference between the two wires, we need to integrate the electric field along any path from one conductor to the other. It is obvious from Figure that one convenient integration path is along the x axis, from $x = a$ to $x = d - a$. We find

$$\begin{aligned} V_{12} &= -\frac{1}{2\pi\epsilon_0} \int_a^{d-a} \left[\frac{-\rho_l}{x} + \frac{+\rho_l}{x-d} \right] dx = \frac{\rho_l}{2\pi\epsilon_0} \int_a^{d-a} \left[\frac{1}{x} + \frac{1}{d-x} \right] dx \\ &= +\frac{\rho_l}{2\pi\epsilon_0} [\ln x - \ln(d-x)]_a^{d-a} = \frac{\rho_l}{2\pi\epsilon_0} \left[\ln \left(\frac{x}{d-x} \right) \right]_a^{d-a} \\ &= \frac{\rho_l}{2\pi\epsilon_0} \left[\ln \left(\frac{d-a}{d-(d-a)} \right) - \ln \left(\frac{a}{d-a} \right) \right] = \frac{\rho_l}{2\pi\epsilon_0} 2 \ln \left(\frac{d-a}{a} \right) \\ &\simeq \frac{\rho_l}{\pi\epsilon_0} \ln \frac{d}{a} \quad (\text{assuming } d \gg a) \end{aligned}$$

Noting that the total charge on one of the conductors of unit length is $Q = \rho_l$, the *per unit length* capacitance C_{pu} of the two-wire line is

$$C_{pu} = \frac{\rho_l}{|V_{12}|} \simeq \frac{\pi\epsilon_0}{\ln(d/a)} \quad \text{F/m}$$



The dashed lines are equipotential lines where flux or field line meet at right angles.

Figure 1: The lines and direction of the electric field in the space between the wires.

Q2: [60] Consider a spherical dielectric shell. A positive point charge Q is at the center of a spherical dielectric shell of an inner radius a and an outer radius b . The relative dielectric constant of the shell is ϵ_r . Determine

1.-3. $\mathbf{E}$, $\mathbf{D}$, and $\mathbf{P}$ every where.

4. The polarization charge densities $\rho_{v\text{pol}}$ and $\rho_{s\text{pol}}$.

5. The electrostatic energy $W_e = \frac{1}{2} \int_V \mathbf{D} \cdot \mathbf{E} dv$.

Solution: Because of the spherical symmetry, we apply Gauss's law to find $\mathbf{E}$ and $\mathbf{D}$ in three regions: (a) $r \geq b$, (b) $a \leq r < b$, and (c) $r < a$. The electric potential Φ is then determined from the negative line integral of $\mathbf{E}$, and polarization $\mathbf{P}$ follows from the relation

$$\mathbf{P} = \mathbf{D} - \epsilon_0 \mathbf{E} = \epsilon_0 (\epsilon_r - 1) \mathbf{E}$$

(a) For the region $r \geq b$ in the free space, $\mathbf{E}$ and V are the same as we know for the point charge. So, we have

$$\mathbf{E} = \frac{Q}{4\pi\epsilon_0 r^2} \mathbf{a}_r \quad \text{and} \quad V = \frac{Q}{4\pi\epsilon_0 r} \quad r \geq b$$

Since we are in the free space, the phenomenon of polarization does not apply, therefore, we have

$$\begin{aligned} \mathbf{P} &= 0 & r &\geq b \\ \mathbf{D} &= \epsilon_0 \mathbf{E} = \frac{Q}{4\pi r^2} \mathbf{a}_r & r &\geq b \end{aligned}$$

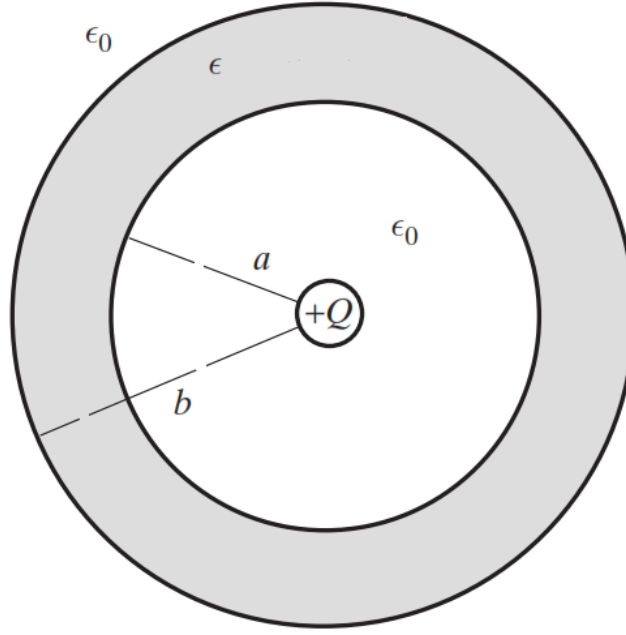


Figure 2: Fig. for Question 2.

(b) For the region $a \leq r < b$ (i.e., inside the dielectric shell), $\mathbf{E}$ is found using Gauss's law as

$$\mathbf{E} = \frac{Q}{4\pi\epsilon_0\epsilon_r r^2} \mathbf{a}_r = \frac{Q}{4\pi\epsilon r^2} \mathbf{a}_r \quad a \leq r < b$$

and $\mathbf{D}$ follows as

$$\mathbf{D} = \epsilon \mathbf{E} = \frac{Q}{4\pi r^2} \mathbf{a}_r \quad a \leq r < b$$

The polarization vector $\mathbf{P}$ is

$$\mathbf{P} = \mathbf{D} - \epsilon_0 \mathbf{E} = \frac{Q}{4\pi r^2} \left(1 - \frac{1}{\epsilon_r}\right) \mathbf{a}_r \quad a \leq r < b$$

(c) For the region $r < a$, the application of Gauss's law yields the same expressions for $\mathbf{E}$, $\mathbf{D}$, and $\mathbf{P}$ as we obtained earlier in the region $r \geq b$. The polarization charge densities associated with the dielectric shell are

$$\rho_{v\text{ pol}} = -\nabla \cdot \mathbf{P} = -\frac{1}{r^2} \frac{\partial}{\partial r} \left[r^2 \frac{Q}{4\pi r^2} \left(1 - \frac{1}{\epsilon_r}\right) \right] = 0$$

inside the dielectric shell. The surface densities of bound/polarized charges are

$$\rho_{s\text{ pol}}|_{r=a} = \mathbf{P} \cdot (-\mathbf{a}_r)|_{r=a} = -\frac{Q}{4\pi a^2} \left(1 - \frac{1}{\epsilon_r}\right)$$

on the inner surface of the shell, and

$$\rho_{s\text{ pol}}|_{r=b} = \mathbf{P} \cdot \mathbf{a}_r|_{r=b} = \frac{Q}{4\pi b^2} \left(1 - \frac{1}{\epsilon_r}\right)$$

on the outer surface of the shell, respectively. The net induced (but bound) charges in the dielectric material must be $-Q$. I leave this for you to prove; ask Vezhish for help.

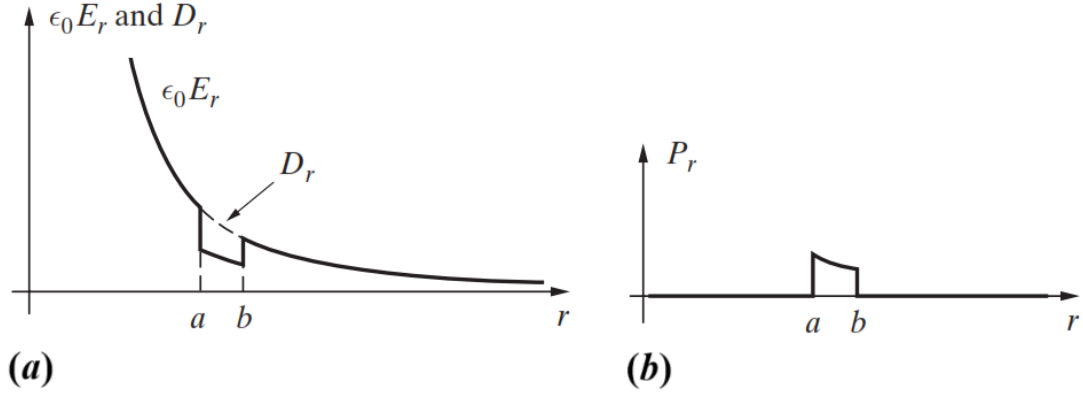


Figure 3: The plots of field, flux density and polarization versus distance from the origin.

Consider a small (non-zero) radius, $r_0 < a$, around the point charge to compute the energy around it. We need to do this because, at $r = 0$, the field expression is singular.

$$\begin{aligned}
 W_e &= \frac{1}{2} \int_{r_0}^{\infty} \mathbf{D} \cdot \mathbf{E} dv = \frac{1}{2} 4\pi \int_{r_0}^{\infty} D_r E_r r^2 dr \\
 &= 2\pi \int_{r_0}^{\infty} D_r E_r r^2 dr = 2\pi \int_{r_0}^a + \int_a^b + \int_b^{\infty} D_r E_r r^2 dr \\
 &= \frac{2\pi Q^2}{(4\pi)^2 \epsilon_0} \int_{r_0}^a + \frac{1}{\epsilon_r} \int_a^b + \int_b^{\infty} \frac{1}{r^2} dr \\
 &= \frac{Q^2}{8\pi \epsilon_0} \int_{r_0}^a + \frac{1}{\epsilon_r} \int_a^b + \int_b^{\infty} \frac{1}{r^2} dr \\
 &= \frac{Q^2}{8\pi \epsilon_0} \left[\left(\frac{1}{r_0} - \frac{1}{a} \right) + \frac{1}{\epsilon_r} \left(\frac{1}{a} - \frac{1}{b} \right) + \left(\frac{1}{b} - \frac{1}{\infty} \right) \right] \\
 &= \frac{Q^2}{8\pi \epsilon_0} \left[\frac{1}{r_0} - \frac{1}{a} + \frac{1}{\epsilon_r} \left(\frac{1}{a} - \frac{1}{b} \right) + \frac{1}{b} \right].
 \end{aligned}$$